

Experimental methodology

MIT-BIH Arrhythmia DB
(48 recordings, 360 Hz)

R-peak 1-s windows, classes N/A/V
per-window normalization to $[-1, 1]$

Patient-level split
train / validation / test (record-aware)

Architectures: CNN . LSTM . GRU . TCN
+ Conv1D autoencoder denoiser

Quantization: FP32 to int8 TFLite
CNN/TCN full-int8; LSTM/GRU float32

Experiments: grouped CV . noise 0-40 dB
SVDB external . temperature calibration

On-device: ESP32-S3 int8 inference,
memory footprint and latency